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Table 1: Analysis Methods

Model	**Input**	**Output**	**What this shows**	**Appropriateness**
XGBoost	Price, Year, Odometer, Manufacturer, Condition, Fuel Type, Cylinder, Region, Demand Score	Predicted Resale Price for the inputted vehicle	Vehicle age, mileage, make, model, specs and last but not least, the local demand is a good indicator of vehicle resale value	It is appropriate for our problem since it works well with complex data and real world relationships. It also is highly accurate and works well on tabular data.
K-Mean Clustering (Vehicle Profile)	Price, Odometer, Year, Fuel Type, Manufacturer, Depreciation Rate	Cluster vehicles with similar depreciation characteristics	K-Mean clustering uses the input features to nearby K points. This helps us identify vehicles with similar depreciation characteristics and overall trend	It is appropriate here because there are underlying natural depreciation patterns that vary across different vehicle make, fuel type and spec. K-Mean is able to separate them and reveal the patterns or groupings naturally
K-Mean Clustering (Region Profile)	Average Depreciation Rate By Region, EV Adoption, Demand Score Per Fuel Type By Region	Cluster CA region with similar depreciation environment	This uncovers depreciation insight across different regions in CA	Similarly, K-Mean is appropriate here because it's able to separate and reveal clusters with similar patterns
KNN	User Inputted Car Profile: Make, Model, Year,	5 or K most similar vehicles	This highlights which vehicles are similar to the user	KNN finds the K most similar neighbors and it suits perfectly for

	Odometer, Fuel Type, Region		inputted vehicle. This serves as a comparable benchmark for the user	this task since it returns 5 most similar vehicles on feature distance
Feature Engineering (Demand Score)	Zipcode, Make, Fuel, Vehicles count from complete_reg; region mapping from ca_vehicles	Demand score per region per make per fuel type	This gives us a demand signal across various regions in CA for various types of vehicle from complete_reg data	The number of registration per region can be used to infer a vehicle's demand at a given region

Table 2: Visualization Plan

Method Name	**X-Axis**	**Y-Axis**	**What Are You Trying To Show**
Histogram(Price)	Price ($) binned ranges	Count of listings	Shows the overall distribution of used car prices in California after cleaning
Boxplot(Price by Manufacturer)	Manufacturer	Price ($)	Compares price distributions across the top 10 most listed brands. Shows median prices, spread and outliers per brand, visually validating the decision to tier vehicles into economy, midrange and luxury categories.
Hexbin (Price vs Odometer)(Used Cars)	Odometer (miles)	Price ($)	Reveals the relationship between mileage and resale price in the used car market. Confirms odometer as a primary depreciation driver and shows where most transactions cluster in the California used car market

Hexbin Price vs Mileage (MSRP Dataset)	Mileage (miles)	Price ($)	Shows the price-mileage relationship for the MSRP dataset. This allow us to analyze how depreciation differs across new and used vehicles
Line Graph (Fuel Type Registration Share Over Time)	Year (2019–2024)	Percentage of total registrations (%)	Tracks how the share of each fuel type changed across California from 2019 to 2024. It shows that battery and electric vehicles grew over 6x while gas vehicles slowly and steadily declined year after year. This supports the hypothesis that gas vehicles depreciate faster in California
Bar Chart (Top 5 Brands by Registration)	Manufacturer	Total registered vehicles	Shows the most registered brands in California over 6 years. Toyota and Honda's stability versus Ford and Chevrolet's notable decline supports the hypothesis that brand demand influences resale value and depreciation rates
Bar Chart (Average Price by Region and Fuel Type)	Region	Average Price ($)	Compares average used car prices across California regions broken down by fuel type. This reveals that electric vehicle prices vary meaningfully by region. This signals geographic depreciation differences across California
Heatmap Feature Correlation Matrix	Features (price, year, odometer)	Features (price, year, odometer)	Visualizes the correlation between core numeric features.

			Confirms that year (0.53) and odometer (-0.55) have the strongest relationships with price.
Scatter Plot (Actual vs Predicted Price)	Actual Price ($)	Predicted Price ($)	Model performance visualization. Points along the diagonal indicate accurate predictions. Deviations reveal where the model struggles, used to evaluate XGBoost model quality and communicate prediction accuracy
Chart (Feature Importance)	Feature names	Importance score	Shows which feature the XGBoost model rely on most when predicting price.